Cryptography Research Framework Blueprint

Primary Goal

Build a long-term modular C++20 cryptography research framework for learning modern C++, Bitcoin internals, image/steganography analysis, benchmarking and future CUDA acceleration. The framework should be generic rather than tied to a single puzzle.

Technology Stack

- C++20
- Visual Studio 2022 Community
- CMake
- vcpkg
- Git
- GoogleTest
- spdlog
- nlohmann/json
- OpenSSL
- libsecp256k1
- OpenCV
- CUDA Toolkit (future phase)

Project Structure

```
CryptoResearchFramework/
├── app/
├── cli/
├── core/
│   ├── crypto/
│   ├── hashing/
│   ├── encoding/
│   └── utils/
├── bitcoin/
```

├── image_analysis/

├── steganography/

├── puzzle_plugins/

├── algorithms/

├── gpu/

├── benchmarks/

├── tests/

├── docs/

├── scripts/

└── third_party/

Architecture Rules

- Modular design.

- No monolithic code.

- New puzzle = new plugin/module.

- Core framework should not require modification when adding new puzzles.

- Prefer RAII, smart pointers and standard library facilities.

- Production-quality code only.

- Cross-platform design (Windows first, Linux compatibility later).

- Unit tests for all core modules.

Development Roadmap

Phase 1: Project setup, SHA-256, RIPEMD-160, Base58Check, Bitcoin address generation.

Phase 2: Image analysis, metadata, OCR, PNG parsing, steganography modules.

Phase 3: Benchmarking, multithreading, performance optimization.

Phase 4: CUDA acceleration, advanced research modules.

Codex Instructions

- Use C++20 best practices.

- Use CMake and vcpkg.

- Avoid placeholder implementations and TODOs.
- Explain non-trivial design decisions briefly in comments.
- Keep every module independent and testable.
- Optimize only after correctness is verified.